School of Computer Science

Applied Machine Learning Project

# Predicting the Results of Professional Tennis Matches with Machine Learning

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Date: 13/05/2026

# Declaration

I hereby declare that the work described in this report is, except where otherwise stated, entirely my own and that I have not submitted it before for summative assessment in third-level education. I have read and understand TU Dublin's policy on plagiarism.

Signed  Date 13/05/2026

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# Dedication

To my wife, Sarah, and my children, Hugo and Oliver, for your love and infinite support.

# Abstract

This project aims to predict winners of professional tennis matches using only pre-match tabular data. Historical tournament, match, ranking and player information from an open-source repository were consolidated into a large dataset of matches spanning 55 years of the ATP Tour, Davis Cup, and the 2024 Olympic singles tournament, and then enriched with several features computed under a strict chronological order, including alternative player ratings based on Elo and Glicko models to capture player strengths and recent form. The dataset was used to train six machine learning classifiers (Random Forest, XGBoost, CatBoost, linear C-Support Vector Classifier, Nyström + Stochastic Gradient Descent, and a Multi-Layer Perceptron), which were then compared using standard performance metrics. The strongest models achieve around 70% accuracy and competitive probability scores, aligned with prior work. Shapley values confirm the significance of the dynamic ratings and player-related features over official ATP ranking positions. An analysis of the results shows how accuracy can vary significantly across years, tournament levels, and match formats.

# Contents

# Acronyms and Abbreviations

| | |
|---|---|
| **Adam** | Adaptive Moment Estimation algorithm to update network weights |
| **ANP** | Analytic Network Process |
| **ATP** | Association of Tennis Professionals |
| **C-SVC** | C-Support Vector Classifier |
| **CatBoost** | an implementation of gradient boosting on decision trees. |
| **CSV** | Comma-Separated Values format for tabular data |
| **DT** | Decision Tree |
| **EDA** | Exploratory Data Analysis |
| **F1** | F1 Score, a metric that balances precision and recall |
| **GBM** | Gradient Boosting Machine |
| **LASSO** | Least Absolute Shrinkage and Selection Operator |
| **LGBM** | Light Gradient Boosting Machine |
| **ML** | Machine Learning |
| **MLP** | Multi-Layer Perceptron |
| **RBF** | Radial Basis Function |
| **ReLU** | Rectified Linear Unit, a non-linear activation function |
| **RF** | Random Forest |
| **ROC-AUC** | Receiver Operating Characteristic - Area Under the Curve |
| **SGD** | Stochastic Gradient Descent |
| **SVM** | Support Vector Machine |
| **WTA** | Women’s Tennis Association |
| **XGBoost** | eXtreme Gradient Boosting algorithm |

# 1 Introduction

Tennis is a popular sport where the one-on-one nature of competition provides an ideal environment for sports analytics, as match outcomes are determined by a more controlled set of variables than in team-based sports. While professional tennis has benefited from rigorous record-keeping and a wealth of statistics about every aspect of the game, accurately predicting match results remains a complex challenge due to the intricate interactions between player skills, form, physical and psychological state, as well as environmental factors such as the weather, court surface type, and even tennis ball characteristics that can significantly influence the performance of players and the outcome of a match.

## 1.1 Project description

The focus of this project is to develop a framework that prioritises the curation of high-quality data and the engineering of domain-driven features that can capture the multiple facets of the game and be used to predict the results of professional tennis matches. Central to this approach is the use of “enhanced covariates”—variables that are not directly observed but are derived through mathematical transformations or separate statistical models.

There has been significant research on developing statistical models to rank tennis players as alternatives to the official rankings published by the professional tennis organisations. In parallel, several studies have applied machine learning to predict the results of tennis matches; however, most of these studies were restricted to specific tournaments (e.g. Grand Slam tournaments only) or did not take full advantage of the available data and statistical models. This project aims to address these limitations by tapping into the vast amount of data available and using machine learning techniques to explore the predictive power of different statistical models proposed in the literature while also discovering new ways to improve the predictions of these models.

## 1.2 Project motivation and justification

Predicting the outcomes of sporting events has been a long-standing problem in the field of sports analytics, and can have significant implications for the development of training methods and strategies to improve the performance of professional athletes. It is also important for broadcasters and the media in general as it helps them to transform a passive viewing experience into an interactive, high-stakes narrative that drives fan engagement and viewer retention. Finally, it can be argued that predicting sporting events can help develop valuable models and gain insights that could be useful in diverse areas of human activity, from financial markets, to politics and international conflicts (Gu and Saaty 2019).

## 1.3 Project scope

This project is focused on predicting the results of professional tennis matches at tournaments on the men’s tour organised by the Association of Tennis Professionals (ATP). While there is a wealth of data available from the Women’s Tennis Association (WTA) as well, limiting the scope to one of the two circuits allows for a more focused and comprehensive analysis in the limited time available for this project. We note that the focus of this project is modelling predictions based solely on pre-match data rather than using data collected while the match is in progress.

Finally, we note that there is significant interest in the literature in using betting odds to either improve predictions or compare their accuracy. While odds that are carefully curated and published by professional bookmakers can be a valuable source of information, this project does not use odds data nor does it focus on this aspect of the problem.

## 1.4 Report layout

The remainder of this report is organised into four main sections after this introduction. Section 2 surveys paired-comparison models and rating systems (including Elo and Glicko), reviews machine-learning studies applied to tennis forecasting, and closes with a review of identified research needs. Section 3 describes the methodology and the end-to-end execution of the work: from the collection of match, ranking and player data to the construction of the dataset and feature engineering, data exploration, and the implementation of several machine learning models. The section concludes with a discussion on how the models were evaluated and some implementation details for reproducibility. Section 4 presents the results, initially comparing the models on a held-out test partition, and then a more comprehensive analysis of accuracy trends by season, tournament level, match format, and court surface. The section concludes with a discussion of the most important features in the dataset and their impact on the models. Section 5 synthesises the empirical findings and discusses the results in the context of the literature, outlining some directions for future work.

# 2 Literature review

The Association of Tennis Professionals (ATP) manages the men's circuit and maintains the official player ranking that is used to determine tournament eligibility, entry qualification, and player seeding. This ranking is based on a cumulative rolling points system. Players earn points for each match won in an official tournament, and points remain on a player's record for 52 weeks before expiring. The number of points awarded for each win is determined by the event category and the round in which the match is played (Williams et al. 2021). For instance, the champion of any of the four annual Grand Slam events (Australian Open, Roland Garros, Wimbledon, and the US Open) receives 2,000 points, whereas the winner of a first-round match in the same tournament who does not advance any further receives 50 points. In contrast, the champion of an ATP 250 event receives only 250 points.

While official rankings are a good indication of the current strengths and abilities of tennis players, and can generally be used to predict the winner of a professional tennis match, they obviously do not capture the full story and are far from being a reliable prediction tool, particularly when the difference in rankings between players is small.

## 2.1 Paired comparisons and rating mechanisms

Noting that official systems often reward frequent participation over absolute performance, researchers set out to develop alternative rating mechanisms that could capture players' abilities by looking at their past match results, particularly taking into account their opponents' strengths rather than the prestige of the tournament or the round in which the match is played.

One of the earliest attempts to develop an improved rating mechanism for tennis was proposed by Glickman (1999), who devised a dynamic paired comparison model to estimate the abilities of players over time. He showed that, given a particular choice of constants, his method is equivalent to the popular Elo rating system developed for chess players by Arpad Elo in 1978. While Glickman used match results to produce an alternative ranking for tennis players, he did not test its forecasting capabilities.

McHale and Morton (2011) pioneered a similar rating mechanism by building a Bradley-Terry type model (a popular model for handling paired comparisons) that utilised players' past match results using an exponential decay function to account for players' recent form (i.e. giving more importance to recent results than older ones). Their model also accounted for the score of the matches. For instance, a player who wins a match with a score of 6–0 6–0 is rewarded more than a player who wins with a score of 7–6 7–6 (presumably, a much tighter match). They also note that tennis is played on different surfaces (e.g. clay, grass, and hard courts) that can significantly change the performance of players, thus they also made further use of weighting techniques so that matches played on surfaces other than that of the tournament being considered are given less importance. They showed that their model provided superior forecasts to the official rankings using data from nine seasons of the ATP tour from 2000-2008, and could provide higher returns against bookmaker odds using a simple betting strategy.

Expanding on this relational logic, Knottenbelt, Spanias, and Madurska (2012) proposed a common-opponent stochastic model that relied on the transitive nature of tennis, predicting match outcomes by calculating the difference in service points won against opponents that both players had previously faced. They argued that the intuition behind this approach is that tennis, to a certain extent, is a transitive sport (i.e. if player A is better than player B and player B is better than player C, then, usually, player A is better than player C). However, as many avid tennis fans will attest, this is often not the case. Using this approach, they were nevertheless able to generate 3.8% long-term profit against the best odds published by bookmakers for a set of over 2,000 tennis matches from the ATP and WTA tours in 2011.

Kovalchik (2016) tested the predictive performance of 11 published forecasting models for predicting the outcome of 2,395 singles matches during the 2014 season. They found that an Elo-based model was the best performing model, achieving an accuracy of up to 70%, although this was still worse than a bookmaker consensus model built by aggregating the odds offered by established bookmakers for each match, which achieved an accuracy of 72%.

Subsequent research heavily explored the Elo rating system. Williams et al. (2021) compared standard Elo, surface-specific Elo ratings, and weighted composites, demonstrating that Elo-based approaches are effective, offering comparable, and sometimes superior, forecasting performance to betting odds on Grand Slam matches between 2018 and 2020. Angelini, Candila, and De Angelis (2022) introduced the concept of a Weighted Elo (WElo), which incorporates the margin of victory (i.e., the number of games or sets won) into the standard Elo updating formula to better reflect players' strengths. They tested their model on a set of matches from the ATP and WTA tours between 2012 and 2020, and showed that the WElo was superior to the standard Elo approach as well as the Bradley-Terry type model of McHale and Morton (2011), among other models.

Fayomi et al. (2022) revisited the Bradley-Terry model, using it to compute win probabilities and new player ratings that demonstrated the profound impact of court surface (especially, clay vs. hard court) on a player's true ability. They applied the model to data from the ATP tour from January 2019 to September 2020, achieving results similar to those of McHale and Morton (2011).

## 2.2 Machine learning approaches

Predictive techniques have transitioned from statistical models to complex machine learning (ML) approaches. An early example is Somboonphokkaphan, Phimoltares, and Lursinsap (2009), who deployed a Multi-Layer Perceptron (MLP)—a foundational class of fully-connected feedforward artificial neural network— that incorporated time-series history, environmental data in the form of surface type, and several statistical features such as winning percentage of first serves and total points won. They tested their model on a set of matches between 2003 and 2008 and reported accuracy rates ranging from 69% to 81%, although it should be noted that these results were restricted to a small set of Grand Slam matches only.

Recognising that historical data alone misses intangible factors, Gu and Saaty (2019) built an Analytic Network Process (ANP)—an advanced, non-linear decision-making framework—combining data analysis with expert human judgments, capturing unquantifiable variables such as a player's psychology, brainpower, and stamina. They reported that their model achieved an accuracy of up to 85.1%; however, once again, this was restricted to a set of only 94 matches from the 2015 US Open tournament. Moreover, this approach required the collection of judgements from tennis experts based on their knowledge and expertise about the matches and the players, an approach that is clearly difficult to scale.

Gao and Kowalczyk (2021) built three machine learning models (Logistic Regression, Random Forest, and Support Vector Machine), performed an in-depth feature selection process, and compiled one of the largest tennis statistics datasets publicly available. They found that a Random Forest classifier achieved the best performance, achieving prediction accuracies exceeding 80% on a set of 49,188 matches from the ATP tour. They also identified serve strength—specifically the percentage of points won on first and second serves—as a key predictor of match outcomes. Wilkens (2021) conducted a comprehensive comparative study using various ML methods (Logistic Regression, Neural Network, Random Forest, Gradient Boosting Machine, and Support Vector Machine) on a dataset of approximately 39,000 matches from the ATP and WTA tours, and noted that differences in prediction performance among the various ML approaches were small.

Yue et al. (2022) applied exploratory data analysis (EDA) to examine variables related to match results and introduced new variables using the Glicko rating system of Glickman (1999), including surface-specific ratings, which were then used to build several statistical and machine learning models (Logistic Regression, Support Vector Machine, Neural Network, and Light Gradient Boosting Machine). The models were trained on a dataset of Grand Slam matches between 2000 and 2019. The results showed only marginal differences in performance between the various models, with the best results achieved using surface-specific Glicko ratings, with up to 73.8% accuracy on the ATP matches.

Buhamra, Groll, and Gerharz (2025) compared several models including regression-based (logistic, LASSO, splines) and machine learning approaches (Classification Tree, Random Forest, linear and non-linear Support Vector Machine, Artificial Neural Network, and XGBoost) on a set of 5,013 Grand Slam matches from 2011 to 2021. Different features were considered including ATP ranking and points, Elo ratings, bookmaker odds, and age variables that take into account the presumably optimal age of a tennis player being between 28 and 32 years old. Moreover, they investigated different forecasting strategies including “expanding window” and “rolling window”. They found that XGBoost achieved the highest classification rates, with up to 83% accuracy with the expanding window strategy.

## 2.3 Identified research needs

Taken together, the studies reviewed in this section illustrate a recurring limitation: conclusions are often drawn from short calendar windows or focus almost exclusively on Grand Slam tournaments. Moreover, as noted by Wilkens (2021), the heterogeneous setup and data used throughout the literature also demand prudence when comparing or even generalising their findings. These shortcomings motivate the emphasis for the present work: assembling and curating a large, multi-decade body of high-quality professional tennis match data, enriching it with derived metrics and several alternative rating formulations (including variants inspired by Elo- and Glicko-type ideas), and, finally, applying multiple machine learning approaches and comparing them under a common protocol, so that differences in predictive performance can be interpreted in light of consistent data and feature engineering.

# 3 Methodology

This section describes how the project was carried out end to end: from the collection and curation of the raw player, ranking and match data, to the engineering of predictive features, and the construction and evaluation of the machine learning models.

## 3.1 Data collection

Historical data from the ATP tour, along with results from Davis Cup events and the 2024 Summer Olympics tennis tournament, was obtained from an open repository maintained by Jeff Sackmann[1], which contains comma-separated (CSV) files of ATP players, rankings and match results. This repository was cloned into a separate fork[2] so that a small number of data-quality issues discovered during experimentation could be corrected in a controlled (and documented) manner. Examples of such fixes include removing duplicate ranking entries, excluding matches in which the placeholder name “Bye” was used instead of a player name (when a player advances directly to the next round of the tournament without playing a match), as well as a few other minor inconsistencies in the source files.

The repository presents information in three main families of CSV files:

(1) **Player files** supply a unique numeric player identifier (ID) together with the player name, playing hand (right, left, ambidextrous, or unknown), date of birth, and height.

(2) **Ranking files** contain the official ATP weekly rankings with ranking date, player ID, ranking position, and ranking points.

(3) **Match files** describe individual completed contests: tournament information (name, level, starting date, court surface, draw size), player IDs for the winner and loser, match format (e.g. best-of-three or best-of-five sets), and the recorded scoreline.

Match files were combined into a single match-level table. Each row was then linked to the player master file on winner and loser IDs so that player attributes could be attached to each match. Ranking files were aligned to matches by selecting the ranking position and points, for each player, at the tournament start date (or the closest available ranking date). Winner and loser player IDs were randomly reassigned as player 1 and player 2, so that the dataset is not biased towards one player or the other. A winner column was added to indicate the winner of the match (1 or 2). Walkovers and retirements were also removed.

## 3.2 Feature engineering

Several features were calculated in a single chronological pass over all matches, with a running state maintained for every player (and for every player–surface pair) so that quantities could be computed from *prior* contests only. In other words, match outcomes were written to the exported row first, and only afterwards were

[1] https://github.com/JeffSackmann/tennis_atp

[2] https://github.com/scarvalhojr/tennis_atp

win/loss counters, head-to-head tallies, and rating updates advanced to include the current result. This guarantees that no statistic leaks information from matches played later in time or from the fixture being predicted.

For each side in a match, the following history-based quantities were derived from official ATP ranking snapshots and from completed matches strictly before the match under consideration:

- **Career-best ranking.** The best (numerically smallest) official ATP singles ranking position the player attained up to the tournament week.
- **Career match volume and win rate.** Total professional matches played to date and the fraction of those matches won.
- **Recent activity and form.** The number of matches played in the last twelve months and the fraction of those matches won. This "year" window is a rolling look-back of 365 days from the tournament start date (rather than strictly calendar months).

Because court surface strongly conditions playing style and results, parallel counters were maintained for each court type on the ATP tour (hard, clay, grass, and carpet). For every player and surface, the same career totals, twelve-month counts and win rates were tracked using only matches played on that surface before the current match. Where the surface was unknown or missing, surface-specific aggregates were left undefined for that row.

**Head-to-head** counts were also attached to each match, recording how many times the first-named player had beaten the second to that date, in two directed views: overall (i.e. all surfaces) and on the current event surface. Symmetric statistics for the opponent were also added.

Alongside the tabular features above, the same chronological scan maintained eight parallel rating scores for every player: four derived from Elo-style models and four from Glicko-style models. In each case the variants are (i) overall versus surface-specific, and (ii) binary win–loss versus *margin-of-victory*-weighted updates. The surface-specific variants are updated only from prior matches played on the same surface, whereas the overall variants are updated from all prior matches regardless of surface. The weighted variants use a margin-of-victory factor to account for the number of games won by the winner and the loser in the match.

**Elo-type ratings.** Every player begins from the conventional initial rating of $1500$. After each completed match, the rating change follows the usual logistic expected-score pairing based on a $400$-point rating scale; the step size is damped as the player's career match count grows, using the scaling factor proposed by Kovalchik (2016):

$$K = \frac{250}{(N+5)^{0.4}} \tag{1}$$

where $N$ is the player's match count prior to the match under consideration. The margin-of-victory factor for the weighted variants is the one proposed by Angelini, Candila, and De Angelis (2022):

$$V = \frac{G_{\text{win}}}{(G_{\text{win}} + G_{\text{lose}})} \tag{2}$$

where $G_{\text{win}}$ and $G_{\text{lose}}$ are total games won in the match by the winner and loser, respectively. This factor is used to update the score of both players from the recorded scoreline. As an example, in a best-of-five sets match, a 6–0 6–0 6–0 score yields $V = 1$, whereas a 0–6 0–6 7–6 7–6 7–6 result yields $V \approx 0.412$.

**Glicko-type ratings.** Glicko ratings use the Glicko-2 variant documented in Glickman (2022), with non-overlapping *rating periods* of $21$ days and system volatility parameter $\tau = 0.5$. In this system, each player has a rating, a rating deviation, and a volatility. The rating deviation measures the uncertainty of the player's rating, and is high when a player is not competing frequently. The volatility indicates the degree of expected fluctuation in a player's rating—it is high when a player has erratic performances and low when the player performs at a consistent level. Matches within a rating period are treated as having occurred simultaneously. The Glicko-2 system has a score factor $S_p$ that can account for the margin of victory. However, unlike in the Elo system, $S_p$ is different for each player $p$ and reflects the score from each player's perspective: for the player who won the match, $S_{\text{winner}} \in (0.5, 1]$, whereas for the player who lost, $S_{\text{loser}} \in [0, 0.5)$ —a draw is denoted by $0.5$, but that is not a valid outcome in tennis. In the unweighted variants, $S_{\text{winner}} = 1$ and $S_{\text{loser}} = 0$ are used, whereas in the weighted variants they are calculated from the margin-of-victory factor $V$ with the following formulas:

$$S_{winner} = 0.5V + 0.5 \tag{3}$$

$$S_{loser} = 1 - S_{winner} \tag{4}$$

Together, each player therefore carries four Elo-based features (`elo`, `welo`, `surface_elo`, `surface_welo`) and four Glicko-based features (`glicko`, `wglicko`, `surface_glicko`, `surface_wglicko`) prior to the match being predicted.

The exported match-level file contains 63 columns. Table 1 lists the full schema with brief descriptions. Features denoted by the placeholder `playerX` stand for *two* CSV columns in each row—one prefixed with `player1` and one with `player2`—carrying the same information for each entrant. All other features appear once per row.

Table 1: Columns in the final dataset CSV.

| **Feature name** | **Description** |
|---|---|
| *Tournament and match fields* | |
| `year` | Calendar year of the tournament start date. |
| `tournament_id` | Unique identifier for the tournament instance. |
| `tournament_start_date` | Scheduled first day of the event (in `YYYY-MM-DD` format). |
| `tournament_name` | Official tournament name from the source files. |
| `tournament_level` | Tournament level (e.g. Grand Slam, Masters 1000, etc.). |
| `surface` | Court surface (e.g. hard, clay, grass, carpet) when known. |

*Continued on next page*

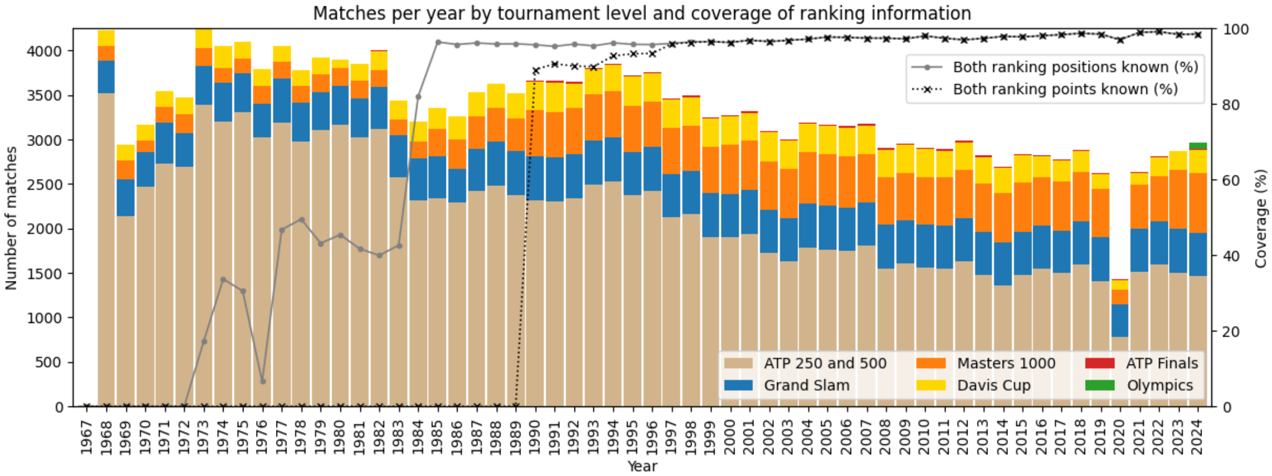
Table 1 — *continued*

| Feature name | Description |
|---|---|
| draw_size | Main-draw size (e.g. 32, 64, 128 players). |
| match_num | Order of the match within the tournament (source ordering). |
| best_of | Match format (e.g. best-of-3 or best-of-5 sets). |
| *Per-player fields (replace X with 1 and 2)* | |
| playerX_id | Unique player identifier. |
| playerX_name | Player's full name. |
| playerX_hand | Playing hand (right, left, ambidextrous, or unknown). |
| playerX_height_cm | Player's height in centimetres when available. |
| playerX_age | Player's age in years at the tournament start date. |
| playerX_rank | Ranking position on the ranking date aligned to the match. |
| playerX_points | Ranking points on the ranking date aligned to the match. |
| playerX_highest_rank | Best ranking position achieved up to the tournament week. |
| playerX_career_matches | Count of prior completed professional matches. |
| playerX_career_win_rate | Fraction of prior professional matches won. |
| playerX_year_matches | Count of matches in the 365 days before the match. |
| playerX_year_win_rate | Fraction of matches won in the 365 days before the match. |
| playerX_surface_career_matches | Prior matches on the same surface type as the event. |
| playerX_surface_career_win_rate | Fraction of matches won on the same surface. |
| playerX_surface_year_matches | Matches on that surface in the 365 days before the match. |
| playerX_surface_year_win_rate | Win rate on that surface in the 365 days before the match. |
| playerX_head2head_wins | Prior wins by this player against the opponent (directed count). |
| playerX_head2head_surface_wins | Prior wins against the opponent on the current match surface. |
| playerX_elo | Overall Elo rating before the match. |
| playerX_welo | Margin-of-victory weighted Elo rating before the match. |
| playerX_surface_elo | Surface-specific Elo rating before the match. |
| playerX_surface_welo | Surface-specific margin-of-victory weighted Elo rating before the match. |
| playerX_glicko | Glicko rating before the match. |
| playerX_wglicko | Margin-of-victory weighted Glicko rating before the match. |
| playerX_surface_glicko | Surface-specific Glicko rating before the match. |
| playerX_surface_wglicko | Surface-specific margin-of-victory weighted Glicko rating before the match. |
| *Outcome* | |
| score | Match scoreline string (sets and tie-break detail). |
| winner | Indicator of which listed player won (1 or 2). |

## 3.3 Data analysis

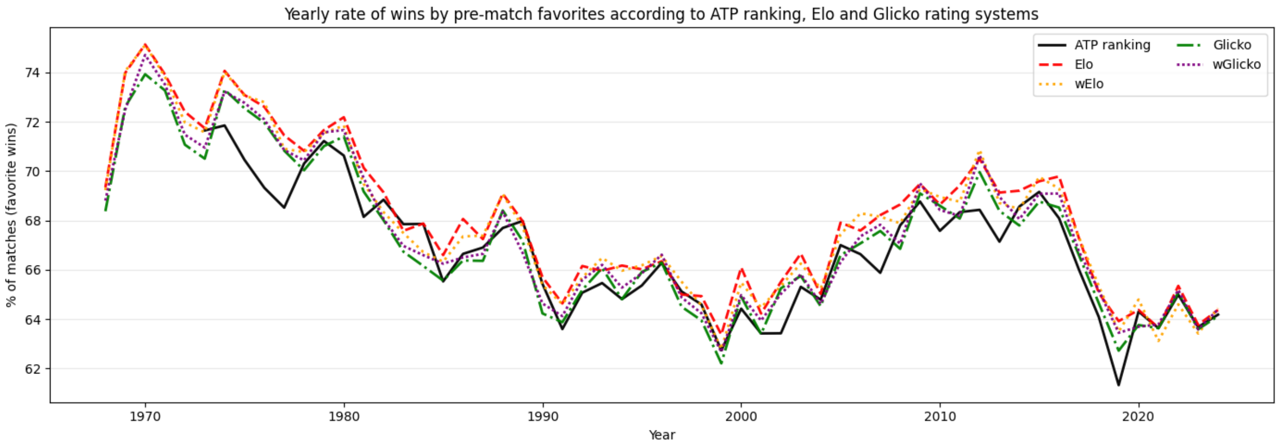
The consolidated dataset resulted in 189,373 matches from 8,454 tournaments spanning the seasons from December 1967 through December 2024, played by 7,477 tennis players. Figure 1 shows the distribution of matches per year by tournament level. The vast majority of matches are from the ATP 250 and 500 tournament levels. The ATP Finals is an annual event that is played by the top-ranked (currently, 8) players of the ATP Tour and, hence, only has a few matches per year. The only year for which data is available from tennis matches played in the Olympics is 2024. Figure 1 also shows the percentage of matches in which the ranking positions and ranking points of both players are known. Ranking positions are lacking in the dataset for a large number of matches before 1985. Similarly, ranking points are missing for all matches before 1990.

Figure 1: Distribution of matches per year by tournament level (on the left y-axis), and the percentage of matches in which the ranking positions and ranking points of both players are known (on the right y-axis).




The predictive power of the official ATP ranking is noted throughout the literature and can be verified with a simple sanity check: among matches where both players have known rankings, define the player with the better (numerically lower) ranking position as the *favourite*; that player's win rate is far above 50% in every season of the dataset. In fact, the same observation applies to the calculated Elo and Glicko ratings (including their weighted and surface-specific variants): treating the higher-rated player before the match as the favourite also yields win percentages well above chance and, hence, these inputs are likely to be powerful candidates for supervised models. Figure 2 shows the yearly share of matches in which the pre-match favourite—by ATP ranking, Elo, weighted Elo (wElo), Glicko, or weighted Glicko (wGlicko)—won the contest. All lines sit clearly above 60% and generally track one another. Surface-specific ratings are not shown but their predictive behaviour is similar. Among these rating systems, the overall Elo is the most accurate on its own, correctly predicting the favourite to win in 68.31% of matches in the dataset, followed by the weighted Elo (68.12%), weighted Glicko (67.73%), and overall Glicko (67.48%). The ATP ranking correctly predicts the winner in 66.24% of matches in which the rankings of both players are known, which is approximately the same as reported by Williams et al. (2021). The surface-specific Elo and Glicko ratings are marginally less accurate than their overall counterparts but are still better than the ATP ranking. Appendix A shows an interesting comparison of the top 30 players according to the official ATP ranking at the end of the 2024 season, together with alternative rankings produced by the Elo and Glicko models.

Figure 2: Yearly share of matches won by the pre-match favourite by ATP ranking versus overall Elo, weighted Elo (wElo), overall Glicko, and weighted Glicko (wGlicko).




## 3.4 Data preparation

After the dataset was assembled, it required little data preparation. Some fields contained occasional missing values and these were addressed with straightforward imputation in the machine-learning pipelines using median or most frequent values.

For supervised experiments, columns were grouped by role. The **target** variable was the match `winner` indicator (1 or 2). **Metadata** columns—carried for traceability but not fed as inputs to the classifiers—consist of tournament information (ID, name, start date, year), match order within the event, the recorded score string, and both players' identifiers and names. **Categorical** inputs comprise match format (`best_of`), tournament level, court surface, and each player's dominant hand; every other predictive column was treated as **numeric** (including rankings, rating histories, head-to-head totals, and derived counts). Categorical features were encoded using one-hot encoding.

The dataset was divided into **training** and **testing** subsets with an $80\,\%$–$20\,\%$ row split, respectively. All models were trained with a grid search to select the best hyperparameters using a 5-fold cross-validation, and the best hyperparameters were then used to refit the model on the full training set. The held-out $20\,\%$ test was then used to report general classification performance.

## 3.5 Supervised Machine Learning Models

The following supervised machine learning models were trained and evaluated using a common framework. Together they span several tree-based classifiers, gradient boosting and bagging techniques, approximate kernel expansions, support vector machines, and a feedforward neural network, many of which appear repeatedly in tennis forecasting studies. Due to the size of the dataset, non-linear support vector machines became computationally infeasible to train.

**Random Forest.** An ensemble of bagged decision trees using *balanced* class weights to mitigate any imbalance between winners. The grid search varies tree depth limits and minimum leaf size. This is a model that has been shown to be effective in tennis forecasting studies and provides a strong, interpretable baseline.

**XGBoost.** An efficient histogram-based gradient-boosted tree classifier, tuned over numbers of boosting rounds, tree depth, learning rate, and row/column subsampling for stochastic regularisation. This is a model that often reaches state-of-the-art accuracy on structured data.

**CatBoost.** Another high-performance gradient-boosted decision-tree implementation, tuned over iterations, depth, and learning rate.

**Calibrated linear SVC.** A Linear C-Support Vector classifier (SVC) wrapped in a Calibrated classifier to produce probabilities. A standard scaler is added to the preprocessing pipeline to ensure features are on a comparable scale. The Linear SVC uses squared-hinge loss and balanced class weights, and is tuned over the regularisation strength. The Calibrated classifier uses a sigmoid calibration with a 5-fold cross-validation to produce win probabilities.

**Nyström + SGD.** The Nyström kernel approximation maps inputs into a Radial Basis Function (RBF) feature space of finite dimension; a linear Stochastic Gradient Descent (SGD) classifier with balanced class weights and a log-loss function is then trained on these features. It also uses a standard scaler in the pipeline. The feature mapping is tuned over kernel scale (gamma) and feature-map dimension (number of components), whereas the classifier is tuned over the regularisation strength.

**Multi-Layer Perceptron.** A fully connected Multi-Layer Perceptron (MLP) classifier using standard scaling preprocessing and the Adam stochastic gradient-based optimiser, with *early stopping* on a held-out fraction of each training fold to limit overfitting. It is tuned over the hidden-layer depth and size, choice of activation function (ReLU, hyperbolic tangent, and logistic), batch size, and regularisation strength.

## 3.6 Model evaluation

All models were assessed on the held-out test subset that played no role in model training or hyperparameter selection. Since the dataset is balanced by construction (for each match, the winner is randomly assigned as player1 or player2), **accuracy** (the fraction of matches classified correctly) is a good metric to assess model performance. Standard metrics such as precision, recall, and the F1 score are also computed using a weighted average over classes. The ROC–AUC is a measure of the model's ability to rank the matches correctly, and is computed as the area under the Receiver Operating Characteristic (ROC) curve. Probabilistic scores complement these point predictions: the **logarithmic loss** (log-loss), measuring the negative log-likelihood of the observed labels under the predicted class probabilities, and the **Brier score**, the mean squared error between the binary outcome and the predicted probability. Together, log-loss and the Brier score judge calibration and sharpness of predicted probabilities, which matters when interpreting forecasts as win chances rather than only as discrete picks. Results are discussed in Section 4.

### 3.7 Implementation details

The processing of the original CSV files into the final match-level dataset with the calculation of all features discussed in Section 3.2 was implemented in Python (code is available online[3]). Elo-style ratings are implemented directly within the codebase. Glicko-2 ratings delegate the update mathematics and period processing to a third-party Python library[4]. The resulting dataset is also available in compressed format online[5].

Supervised machine learning models were implemented using the scikit-learn framework[6], including the built-in implementations of Random Forest, calibrated linear SVC, Nyström kernel approximation, Stochastic Gradient Descent, and Multi-Layer Perceptron, alongside third-party libraries for the XGBoost[7] and CatBoost[8] implementations. Full details of the implementations together with the hyperparameters are available in a Jupyter Notebook[9].

[3] https://github.com/scarvalhojr/tennispredictor/tree/main/code/databuilder
[4] https://pypi.org/project/glicko2-py/
[5] https://github.com/scarvalhojr/tennispredictor/blob/main/data/atp_matches.csv.gz
[6] https://scikit-learn.org/
[7] https://xgboost.ai/
[8] https://catboost.ai/
[9] https://github.com/scarvalhojr/tennispredictor/blob/main/notebook/tennispredictor.ipynb

# 4 Results

This section presents a quantitative analysis of the supervised machine learning models described in Section 3.5. It begins with a direct comparison of the models on the held-out test subset from Section 3.4, using classification and probability scores. The models are then retrained on the full dataset so as to study their predictive performance over time and across different match contexts, before calculating feature importance for the strongest model in order to relate predictive gains to specific engineered inputs.

## 4.1 Model performance on the held-out test subset

After the grid search found the best hyperparameters for each model, all models were evaluated on the same held-out test subset of $37{,}875$ matches (20% of the full dataset), as explained in Section 3.4. Table 2 lists the results with several performance metrics: absolute accuracy; weighted precision, recall and F1 score; ROC–AUC; Brier score; and log-loss.

Table 2: Classification and probability scores on the held-out test subset ordered by accuracy.

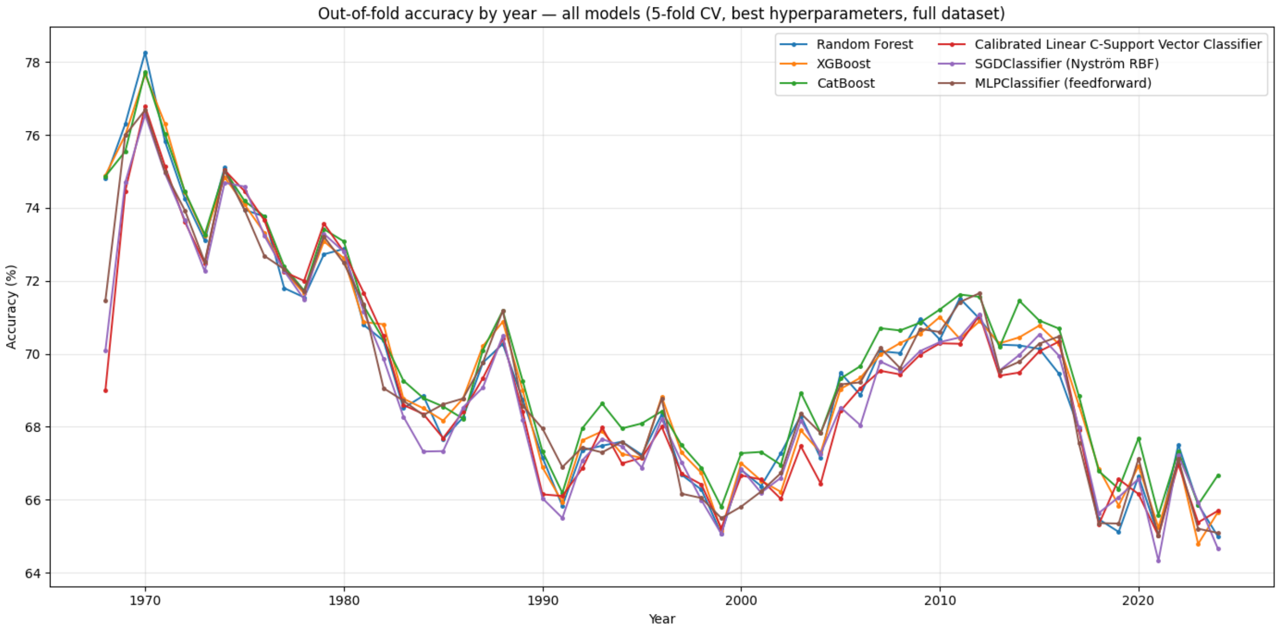
| Model | Accuracy | Precision | Recall | F1 | ROC–AUC | Brier | Log-loss |
|---|---|---|---|---|---|---|---|
| CatBoost | 0.6997 | 0.6998 | 0.6997 | 0.6997 | 0.7723 | 0.1941 | 0.5687 |
| XGBoost | 0.6985 | 0.6985 | 0.6985 | 0.6985 | 0.7716 | 0.1944 | 0.5696 |
| MLP | 0.6953 | 0.6954 | 0.6953 | 0.6953 | 0.7677 | 0.1959 | 0.5728 |
| Random Forest | 0.6951 | 0.6951 | 0.6951 | 0.6951 | 0.7674 | 0.1961 | 0.5738 |
| Nyström + SGD | 0.6923 | 0.6927 | 0.6923 | 0.6923 | 0.7620 | 0.1984 | 0.5799 |
| Linear SVC | 0.6898 | 0.6898 | 0.6898 | 0.6898 | 0.7628 | 0.1979 | 0.5783 |

Differences between models are only modest: gradient-boosted trees (CatBoost, then XGBoost) achieve the highest accuracy and best probability scores (lowest Brier and log-loss). The other models trail slightly but remain competitive, suggesting that much of the predictive signal is already captured by the engineered tabular features rather than by a particular inductive bias alone. Despite several efforts to tune the hyperparameters of the models, they all seemed to hit a ceiling of accuracy around 70%, which is in line with observations in the literature for similar datasets, in particular those reported by Williams et al. (2021). The fact that different machine learning models perform similarly was also noted by Williams et al. (ibid.) and Yue et al. (2022). In general, the results are satisfactory given that most of the features are somewhat related to one another—e.g. players with higher ATP rankings tend to have higher Elo and Glicko ratings, higher win rates, and so on. The Brier scores of all models were below 0.2, and the log-loss was below 0.58, both of which can be considered good scores and are comparable with most of the results reported in the literature.

## 4.2 Retraining models on the full dataset

In order to do a more comprehensive comparison of the classification performance of all models, they were retrained on the full dataset of 189,373 matches using a 5-fold cross-validation approach (with shuffling) using the best hyperparameters found in the previous section. Figure 3 shows the accuracy of predictions by year from all models on the full dataset. The chart shows that all models perform similarly, and that the accuracy of predictions varies between years in a way that resembles how the predictive power of the ATP ranking, Elo and Glicko ratings were shown to vary over time (in Figure 2). The significant variation in accuracy over time also suggests that datasets restricted to specific years can produce vastly different results, which may explain some of the better performing models reported in the literature.

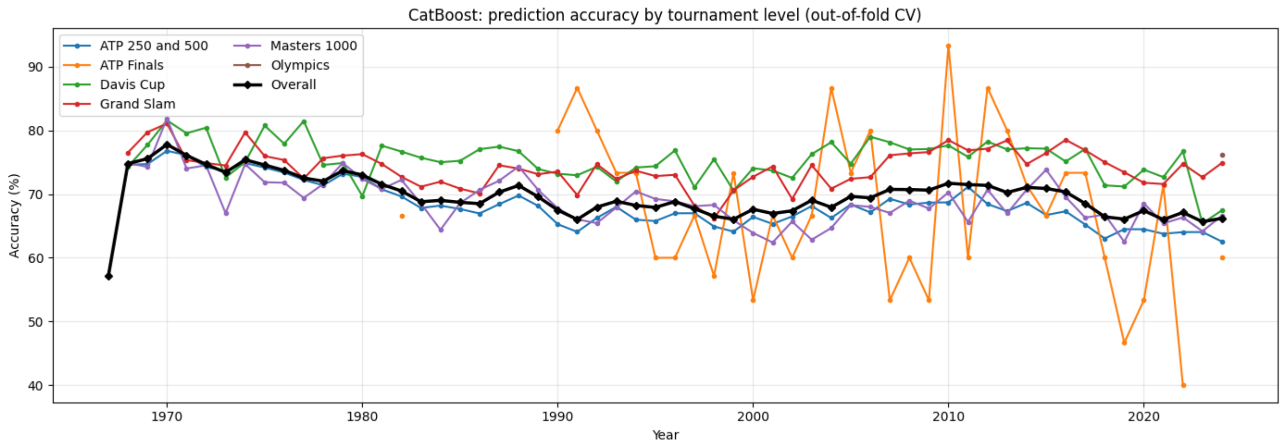
Figure 3: Overall accuracy of predictions by year from all models after re-training on the full dataset with a 5-fold cross-validation.




Given that many results reported in the literature were restricted to Grand Slam matches, it is interesting to see how the models reported here perform on this type of tournament. Figure 4 shows the accuracy by year of the best performing model (CatBoost) for different tournament levels (all other models produced similar results). It is clear that the accuracy is higher on Grand Slam matches and also on Davis Cup matches. This may be explained by the fact that, unlike other tournaments on the ATP Tour, all Grand Slam matches and a large proportion of Davis Cup matches prior to 2018 were played in best-of-five sets format, which is less likely to result in the favourite player losing the match (the underdog needs to win three sets, rather than just two, against the favourite player). Another factor might also contribute to the higher accuracy on Davis Cup matches: many matches in the earlier rounds of the competition tend to involve much lower-ranked players from less established nations in the tennis circuit, so often the favourite player is far superior to the opponent. Finally, the wider variation in accuracy for the ATP Finals is due to the fact that this tournament only has a few matches per year, so the accuracy is more susceptible to random variation.

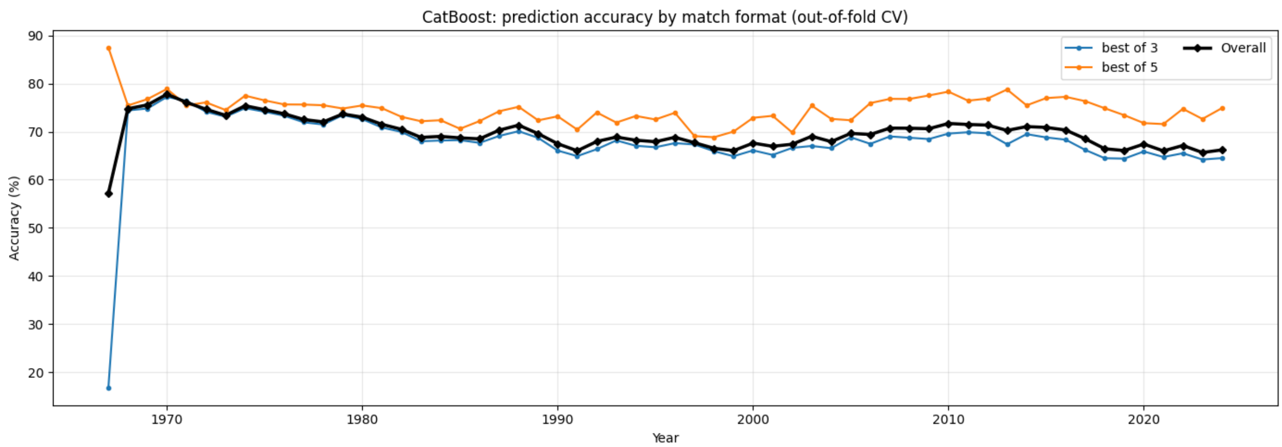
Figure 5 confirms the generally higher accuracy of predictions on matches played in best-of-five versus best-of-three format (results from the CatBoost model only, but all other models produced similar results). Finally, Figure 6 (again, results from the CatBoost model only) does not show any significant difference in accuracy between matches played on different court surfaces, other than perhaps a slightly wider range of accuracy on

Figure 4: Accuracy of CatBoost predictions by year and tournament level.




grass courts that can be explained by the fact that there are fewer tournaments on grass than on other surfaces in a calendar year.

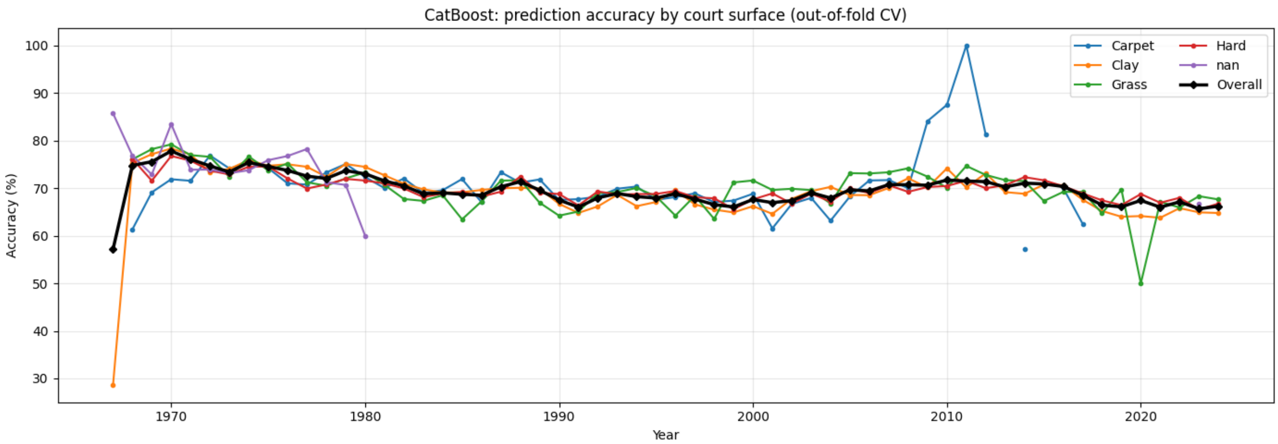
Figure 5: Accuracy of CatBoost predictions by year and match format.




## 4.3 Feature importance

Shapley values were calculated for features within each model—these values are a measure of the contribution of each feature to the prediction of a model as well as the directionality of the impact they have on the predictions. They are calculated by subtracting the average prediction of the model from the prediction of the model when the feature is set to its mean value. Figure 7 shows the top features in the dataset and how they impact the output of the CatBoost model. Features are ordered so the most important features appear at the top. It is interesting to see that the 8 Elo and 7 Glicko features appear among the most important features, and most are considered more important than the ATP ranking positions. The ATP ranking points as well as head-to-head tallies do not appear in the plot so they are considered less important by the model. Other important features are the number of matches played (overall and surface-specific, career and last year counts).

Figure 6: Accuracy of CatBoost predictions by year and court surface.




Somewhat surprisingly, the players' ages appear quite prominently in the plot.

Figure 7: A beeswarm plot showing the most important features in a dataset and their impact on the CatBoost model output, ordered by their Shapley values (most important first). Colours indicate the feature values, with higher values in red and lower values in blue. The direction of the impact they have on the predictions is also indicated, with player 2 winning to the left and player 1 winning to the right.

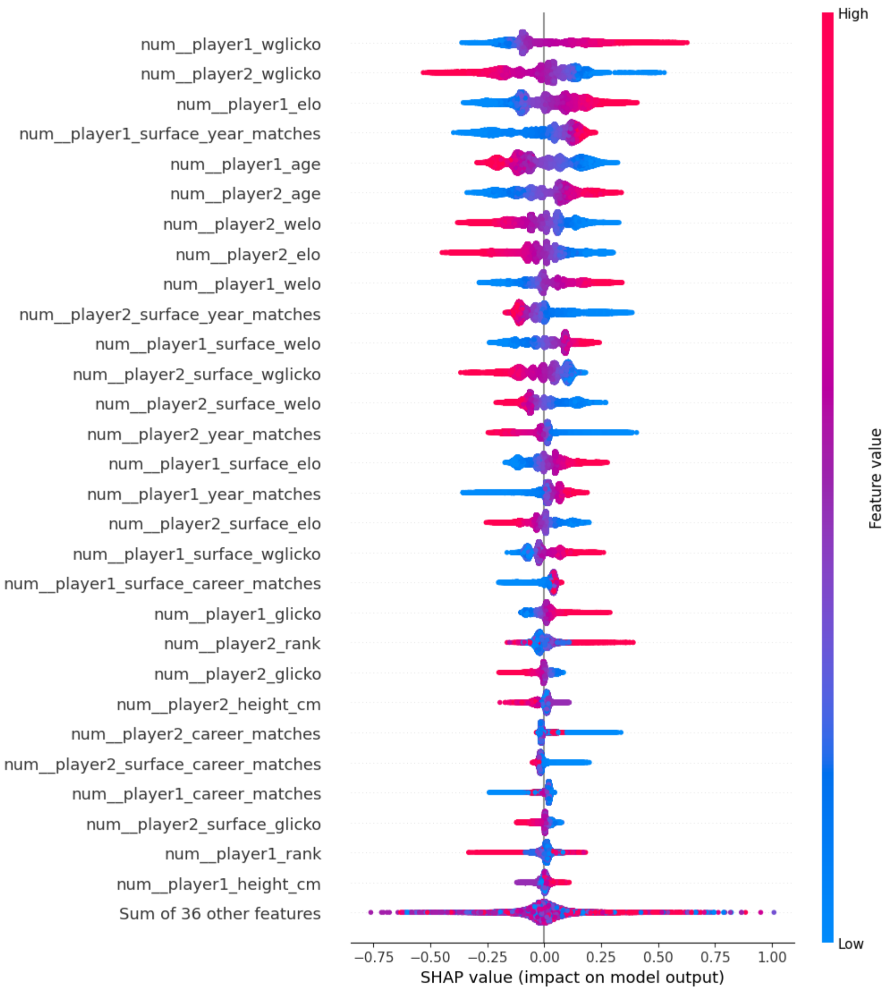
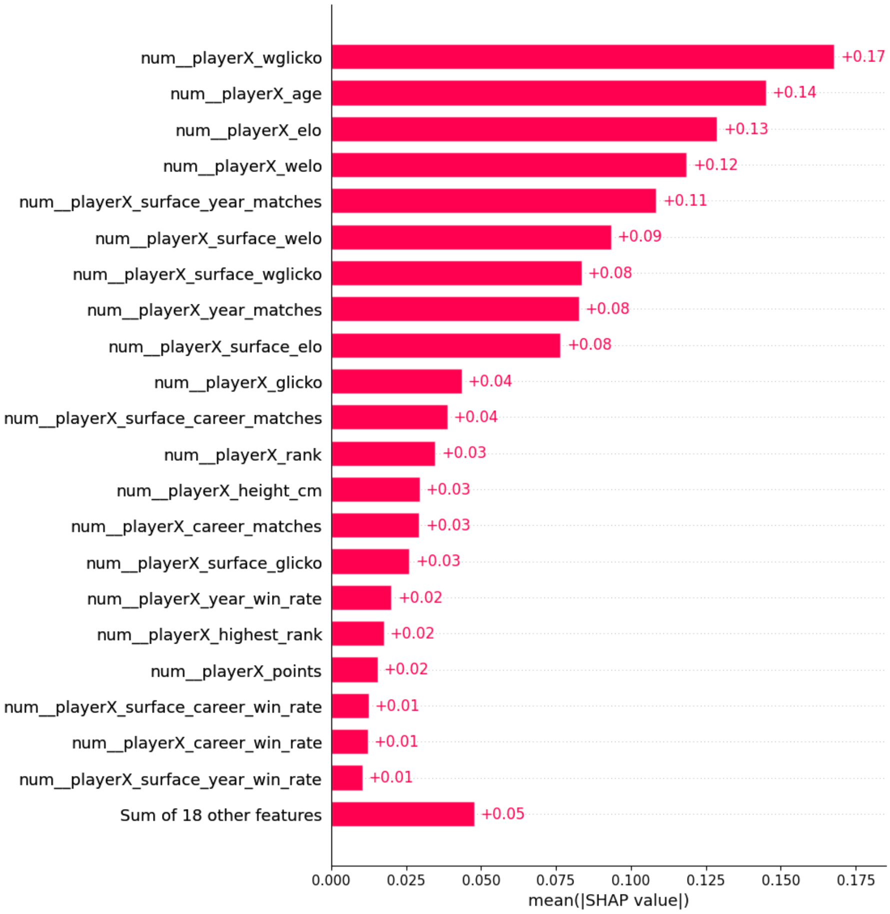


Figure 8 shows a different view of the Shapley values, grouping symmetrical features from both players (`player1` and `player2`) into a single value (`playerX`), once again ordered by the group's combined Shapley value, with the most important ones appearing at the top. The plot largely confirms the findings of the previous plot, although some changes are noticeable; for instance, the players' ages are even higher in the grouped plot.

Figure 8: A bar chart showing the Shapley values of the top grouped features in a dataset, indicating their contribution to the CatBoost model output (most important features first).



# 5 Conclusion

Predicting the winner of a professional tennis match is difficult in a way that no tabular machine-learning model can fully “solve”. Many decisive factors are either unobserved in historical scorelines or only weakly proxied by engineered features: weather (wind, temperature, humidity or rain), day-to-day variation in court pace and ball bounce, the mental and physical state of each player (fatigue, lack of match rhythm, off-court stressors, lack of sleep, injury or illness), mid-match interruptions (such as rain delays), congested tournament schedules, travel disruptions (such as flight delays or cancellations), and the intangible pressure of the occasion (such as the pressure of a final, or the pressure of a home crowd). A vivid illustration is Rafael Nadal’s fourth-round loss to Robin Söderling at Roland Garros in 2009: Nadal had won the title in each of the four preceding years and in fact had never lost a match on the Parisian clay. Moreover, he had beaten Söderling 6–1, 6–0, on clay, only weeks earlier in Rome—yet the result still went against him on the day. No model trained on conventional pre-match statistics could have been expected to assign high probability to that upset while remaining well calibrated across decades of tour play.

Against that backdrop, this project nonetheless produced relatively strong supervised classifiers on the limited historical information available, and broadly replicated the headline performance levels reported in much of the prior literature, with the best models reaching accuracy on the order of 70% on a large, heterogeneous sample of ATP-level match data. At a more granular level, the experiments highlighted a recurring pattern: gradient-boosted tree ensembles (CatBoost and XGBoost) slightly edged other model families on both discrete accuracy and probability scores. Moreover, alternative rating systems (Elo and Glicko) together with several variants were found to be better at predicting player strengths than the official ATP ranking, improving the accuracy of the models as confirmed by the feature importance analysis with Shapley values. One important observation is that, while surface-specific Elo and Glicko ratings can help improve the models, the non-surface-specific variants proved more relevant for predicting the winner of a match. Finally, the fact that different machine learning techniques performed similarly suggests that much of the predictive signal has been captured by the engineered features rather than by model-specific inductive biases.

The work also assembled what is, to our knowledge, the largest consolidated match-level dataset used in this strand of research, spanning over 55 years across several tournament levels. That breadth matters for interpretation: several higher headline accuracies in the literature may reflect evaluation on narrower slices of history and/or on subsets dominated by Grand Slam matches, where best-of-five formats make favourites less vulnerable to single-set volatility than in other tournaments. In application terms, the models are therefore best understood as robust baselines for structured forecasting on archival statistics, not as substitutes for domain expertise when rare contextual shocks can dominate the narrative of a match.

## 5.1 Future work

A natural direction for future work is to enrich the feature space with any additional information available (some of which are listed below) or other external data sources that could help capture some of the missing factors above (such as court surface speed indices, detailed scheduling and travel metadata, injury reports, etc.). Other areas for further improvements include calibrating models for different eras (taking into account the variation in the predictive power of certain features over time), and exploring different training and evaluation strategies to improve and assess the robustness of the models—for instance, by using time-aware training-testing splits with "sliding" or "expanding" windows as described in Buhamra, Groll, and Gerharz (2025).

Below are some of the more readily available features and data that could be added to the dataset, either directly from the same source repository used in this project, derived from it, or from other related sources. Some of the features available from the source repository were not included in the dataset simply because they were not accurately reported or they were not available for a large number of matches, which would require a more complex preprocessing pipeline to handle.

**Challenger, Futures and Qualifying matches.** In this study, matches from the lower levels of the ATP circuit, such as the Future and Challenger events, as well as the matches from the qualifying rounds of the main level tournaments, were not included in the dataset even though they are available in the source repository. This is because these matches feature much lower-ranked players, which require a more careful preprocessing, particularly to calibrate the Elo and Glicko ratings in relation to higher-ranked players.

**WTA matches.** A similar repository from the same author exists for the Women's Tennis Association (WTA) circuit[10], which could be used to augment the dataset and train models taking into account the different characteristics of female tennis players.

**Historical match-level statistics.** Several statistics are available in the source repository about each match, such as the number of aces, double faults, first serves made, break points faced and won, etc. These statistics have the potential to augment the information about strengths of players and their recent form, although it is likely that most of that signal is already captured by other features in the dataset, particularly the Elo and Glicko ratings.

**Tournament round the match was played in.** The tournament round the match was played in is also available in the source repository and could play a role in the predictions in several ways, reflecting how differently some players sometimes perform in certain rounds of a tournament. For instance, players may underperform in the first rounds due to lack of rhythm and match fitness. Others may feel the pressure of a final, particularly the first time they reach that stage of a tournament.

**Player's entry into the tournament.** Some players enter the tournament directly with their ranking, whereas others have to play in the qualifying rounds before they can gain entry to the main draw, which may give them an edge due to the additional time playing and getting to know the exact conditions of the courts. Others may receive a "wild card" by the tournament organisers, often reserved to lower-ranked players such as promising young talents, or former stars who have been out of competition for a while. Players returning from injury may enter the tournament using "protected rankings", which may mean they are lacking match fitness. Finally, there is the case of the "lucky loser", a player who failed to qualify for the main draw but gained a last-minute entry due to another player withdrawing from the tournament—these are notoriously dangerous players who, having already accepted their departure from the tournament, play with immense freedom and zero pressure when given a second chance.

[10] https://github.com/scarvalhojr/tennis_wta

**Total time on court.** The source repository also lists the total time of each match, which could be a useful tool to assess the fatigue of players, the impact of long matches on their performance, or the lack of match rhythm. For instance, it is not uncommon for players to lose in the first round of a tournament immediately after winning the previous tournament where they had to play several long matches.

**Home advantage.** Many players benefit from the support of the crowd when they play in their home country, while others underperform when playing in front of a hostile crowd, hence, knowing the player's home country and where the tournament is played could help improve the predictions.

**Specific court conditions.** While the court surface was used as a feature in the dataset, there can be significant variation in the speed and ball bounce between courts of different tournaments, which can benefit or hinder certain players. For instance, the clay courts of the Madrid Open, which is played at an altitude of 660 metres above sea level, are known to be a lot faster than the clay courts of the Munich Open, which is played at a lower altitude and often in much colder and wetter conditions. Hard courts also have different characteristics, with some being faster or slower than others. Even closing the roof of certain courts (such as the Arthur Ashe Stadium at the US Open) can have an impact on how certain players perform.

**Past performance in the same tournament.** Certain players often perform better (or worse) in certain tournaments (sometimes due to some of the reasons listed above). Knowing how a player performed in a tournament in previous years is often a good indicator of how they will perform the next time they play that tournament again.

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# A Ranking comparisons

Table 3 and Table 4 show the top 30 players by the official ATP ranking at the end of the 2024 season, with alternative rankings produced by several Elo and Glicko scores (weighted and non-weighted, surface-specific and overall).

Table 3: ATP ranking with the top 30 players at the end of the 2024 season, compared with alternative rankings produced by Elo, Glicko, weighted Elo (wElo) and weighted Glicko (wGlicko) models; for each model, two rankings are shown: one updated with all matches in the dataset (no prefix), and one updated with hard court matches only (prefixed with “h-”).

| ATP | Name | Elo | wElo | Glicko | wGlicko | h-Elo | h-wElo | h-Glicko | h-wGlicko |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Jannik Sinner | 1 | 1 | 2 | 1 | 1 | 1 | 2 | 2 |
| 2 | Alexander Zverev | 5 | 5 | 6 | 5 | 6 | 6 | 6 | 7 |
| 3 | Carlos Alcaraz | 3 | 4 | 3 | 3 | 3 | 4 | 3 | 4 |
| 4 | Taylor Fritz | 7 | 7 | 10 | 9 | 7 | 7 | 13 | 13 |
| 5 | Daniil Medvedev | 6 | 6 | 5 | 4 | 5 | 5 | 4 | 5 |
| 6 | Casper Ruud | 28 | 21 | 23 | 21 | 37 | 31 | 42 | 40 |
| 7 | Novak Djokovic | 2 | 2 | 1 | 2 | 2 | 2 | 1 | 1 |
| 8 | Andrey Rublev | 22 | 17 | 16 | 14 | 18 | 12 | 12 | 14 |
| 9 | Alex De Minaur | 11 | 10 | 9 | 8 | 12 | 11 | 9 | 12 |
| 10 | Grigor Dimitrov | 9 | 8 | 8 | 7 | 8 | 9 | 8 | 11 |
| 11 | Stefanos Tsitsipas | 16 | 12 | 13 | 11 | 23 | 17 | 21 | 21 |
| 12 | Tommy Paul | 15 | 15 | 18 | 16 | 20 | 20 | 24 | 24 |
| 13 | Holger Rune | 13 | 16 | 15 | 17 | 13 | 18 | 18 | 22 |
| 14 | Ugo Humbert | 21 | 23 | 26 | 23 | 11 | 13 | 19 | 17 |
| 15 | Jack Draper | 8 | 9 | 12 | 10 | 9 | 8 | 10 | 10 |
| 16 | Hubert Hurkacz | 12 | 13 | 14 | 13 | 17 | 16 | 17 | 19 |
| 17 | Lorenzo Musetti | 29 | 30 | 27 | 29 | 69 | 78 | 90 | 101 |
| 18 | Frances Tiafoe | 35 | 35 | 34 | 38 | 31 | 34 | 40 | 47 |
| 19 | Karen Khachanov | 18 | 22 | 22 | 22 | 19 | 22 | 25 | 28 |
| 20 | Arthur Fils | 25 | 33 | 39 | 40 | 27 | 35 | 44 | 43 |
| 21 | Ben Shelton | 27 | 26 | 30 | 30 | 24 | 23 | 27 | 32 |
| 22 | Sebastian Korda | 17 | 20 | 19 | 18 | 21 | 21 | 22 | 25 |
| 23 | Alejandro Tabilo | 59 | 61 | 65 | 61 | 100 | 102 | 145 | 132 |
| 24 | Alexei Popyrin | 30 | 41 | 49 | 52 | 28 | 41 | 55 | 56 |
| 25 | Tomas Machac | 10 | 19 | 20 | 20 | 10 | 19 | 23 | 27 |
| 26 | Jordan Thompson | 33 | 40 | 41 | 46 | 26 | 33 | 41 | 45 |
| 27 | Sebastian Baez | 79 | 71 | 79 | 87 | 159 | 200 | 230 | 1852 |
| 28 | Jiri Lehecka | 31 | 29 | 31 | 35 | 38 | 40 | 45 | 54 |
| 29 | Felix Auger Aliassime | 34 | 25 | 28 | 27 | 43 | 29 | 36 | 36 |
| 30 | Francisco Cerundolo | 42 | 42 | 47 | 50 | 55 | 64 | 89 | 99 |

Table 4: ATP ranking with the top 30 players at the end of the 2024 season, compared with alternative rankings produced by surface-specific Elo, Glicko, weighted Elo (wElo) and weighted Glicko (wGlicko) models; for each model, two rankings are shown: one updated with clay court matches only (prefixed with “c-”), and one updated with grass court matches only (prefixed with “g-”).

| ATP | Name | c-Elo | c-wElo | c-Glicko | c-wGlicko | g-Elo | g-wElo | g-Glicko | g-wGlicko |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Jannik Sinner | 7 | 8 | 9 | 7 | 4 | 6 | 5 | 5 |
| 2 | Alexander Zverev | 6 | 5 | 6 | 5 | 13 | 14 | 13 | 14 |
| 3 | Carlos Alcaraz | 3 | 3 | 2 | 2 | 2 | 3 | 3 | 3 |
| 4 | Taylor Fritz | 18 | 15 | 20 | 16 | 14 | 17 | 37 | 37 |
| 5 | Daniil Medvedev | 12 | 18 | 15 | 12 | 7 | 8 | 16 | 19 |
| 6 | Casper Ruud | 8 | 7 | 10 | 8 | 1984 | 1974 | 1979 | 1976 |
| 7 | Novak Djokovic | 2 | 2 | 1 | 1 | 1 | 1 | 1 | 1 |
| 8 | Andrey Rublev | 13 | 10 | 11 | 9 | 20 | 20 | 15 | 18 |
| 9 | Alex De Minaur | 19 | 27 | 33 | 32 | 11 | 11 | 23 | 25 |
| 10 | Grigor Dimitrov | 22 | 16 | 18 | 15 | 19 | 16 | 30 | 27 |
| 11 | Stefanos Tsitsipas | 4 | 4 | 4 | 3 | 39 | 39 | 44 | 44 |
| 12 | Tommy Paul | 27 | 34 | 52 | 40 | 8 | 9 | 24 | 23 |
| 13 | Holger Rune | 11 | 13 | 16 | 19 | 22 | 33 | 54 | 64 |
| 14 | Ugo Humbert | 178 | 203 | 214 | 194 | 37 | 34 | 57 | 54 |
| 15 | Jack Draper | 82 | 80 | 65 | 58 | 16 | 18 | 8 | 7 |
| 16 | Hubert Hurkacz | 16 | 24 | 27 | 27 | 9 | 10 | 9 | 11 |
| 17 | Lorenzo Musetti | 9 | 12 | 13 | 10 | 12 | 19 | 51 | 57 |
| 18 | Frances Tiafoe | 72 | 71 | 93 | 96 | 23 | 27 | 49 | 45 |
| 19 | Karen Khachanov | 15 | 14 | 17 | 18 | 43 | 30 | 33 | 35 |
| 20 | Arthur Fils | 39 | 48 | 43 | 45 | 62 | 76 | 1990 | 1986 |
| 21 | Ben Shelton | 99 | 112 | 124 | 138 | 91 | 106 | 2108 | 2091 |
| 22 | Sebastian Korda | 50 | 56 | 59 | 60 | 21 | 21 | 11 | 13 |
| 23 | Alejandro Tabilo | 62 | 53 | 68 | 54 | 32 | 44 | 36 | 47 |
| 24 | Alexei Popyrin | 75 | 90 | 73 | 92 | 1983 | 2039 | 2058 | 2031 |
| 25 | Tomas Machac | 28 | 46 | 28 | 36 | 1998 | 2030 | 2107 | 2110 |
| 26 | Jordan Thompson | 277 | 2079 | 266 | 1979 | 35 | 38 | 52 | 49 |
| 27 | Sebastian Baez | 24 | 20 | 23 | 22 | 2105 | 2099 | 2047 | 2045 |
| 28 | Jiri Lehecka | 36 | 57 | 39 | 49 | 34 | 50 | 22 | 30 |
| 29 | Felix Auger Aliassime | 20 | 25 | 29 | 26 | 40 | 24 | 14 | 16 |
| 30 | Francisco Cerundolo | 21 | 21 | 24 | 23 | 83 | 92 | 80 | 1945 |